

QUANTUM COMPUTING**INTRODUCTION**

Today's computers, both in theory and practice (PCs, laptops, tablets, smartphones...,) are based on classical physics, hence they are called classical systems. Their design is based on the semiconductor technology, but the physics of semiconductor is based on the principles of quantum mechanics. The present day computers rely on very large scale integrated circuits (VLSI), where the important component is the silicon chip of area 0.25mm thick and surface area of 1cm². This contains millions of transistors which act as logic gates. The present trend of semiconductor technology is to bring in more and more transistors onto the silicon chip and increase the clock speed of the microprocessor as per Moore's law. I.e., there will be decrease in the number of electrons in a transistor / device.

Moore's law states that – "In every two years the density of transistors per silicon chip doubles"

But, a device of such microscopic dimensions may not behave the same way as a device of bigger dimensions. Due to this physical limitations, new chip architectures and materials will be used to develop new types of computing that will promote future technological gains. Hence Moore's law no longer is applicable. This analogy of the relationship between the physics and geometry of the materials will help the paradigm shift in computation. Hence the study of information storage, processing and computation can be linked to the study of physical devices which stores the information. This means one must know the Physics principles and material science to design the computers and its computations.

As quantum mechanics underlies the working of traditional, classical computers and communication devices, from the transistor through the laser to the latest hardware advances that increase the speed and power and decrease the size of computer and communications components. Quantum computers and its networks are more efficient than the classical computers in performing the computational tasks such as factorizing the numbers or carrying out searches in a data base.

Just as classical computers can be thought of in Boolean algebra terms, quantum computers are reasoned about with quantum mechanics.

QUANTUM COMPUTING

Quantum computing is a combination of quantum physics, computer science, and information theory. It is the field that investigates the computational power and other properties of computers based on quantum-mechanical principles. It combines two of the most important trends of 20th-century science: quantum mechanics and computer science. Its important objective is to find quantum algorithms that are significantly faster than any classical algorithm solving the same problem.

In Quantum computers, *information is stored in and manipulated using single isolated atoms or photons or electrons.* Since atoms, electrons and photons obey the principles / laws of quantum mechanics, the quantum computers perform computations and process information according to the principles of quantum mechanics.

Richard P Feynman told that the simulations of the quantum systems cannot be done efficiently in the present classical computers. He found that to simulate the quantum systems using classical computers, *the number of computational steps raises exponentially both with the size of quantum system and with the amount of time over which the systems behavior is tracked*. In classical systems the computations and information process will be done by the *ensembles of atoms*. i.e., they are limited by locality i.e., by the classical fact that systems can be in only *one state at the time*.

A quantum system can be in a *superposition of many different states at the same time*, and can exhibit *interference effects* during the course of its evolution. Spatially separated quantum systems may be *entangled* with each other and operations may have “*non-local*” effects because of this.

Quantum computing is not about changing the physical substrate on which computation is done from classical to quantum, but rather changing the notion of computation itself. The change starts at the most basic level: the fundamental unit of computation is no longer the bit, but rather the *quantum bit or qubit*. Placing computation on a quantum mechanical foundation led to the discovery of faster algorithms, novel cryptographic mechanisms, and improved communication protocols.

CLASSICAL (Vs) QUANTUM COMPUTING

General Information:

	Classical Computers	Quantum Computers
Fundamental Unit	Bit	Qubit
Gates	Logic Gates	Unitary Gates
Gates Reversible	Sometimes	Always
Universal Gate Set	NAND	CNOT
Algebra	Boolean	Linear
Definition	Describes the motion of macroscopic objects.	Describes the nature of object at the atomic level.
Principle	If we know the initial state of object we can predict its future state and also can account for past state.	Energy, momentum, angular momentum are quantized.
Nature	Does not describe wave-particle duality	Describes wave-particle duality.

Differences:

Classical Computers	Quantum Computers
It is a large scale integrated multipurpose computer.	It is a high speed parallel computer based on quantum mechanics.
Information storage is Bit based on charge, voltage etc.,	Information storage is quantum bit based on the direction of electron spin.
Discrete number of possible states: 0 or 1	Infinite, continuous number of possible states.
Power increases in a 1:1 relationship with number of transistors.	Power increases exponentially with the number of qubits.

Information / Data processing is carried out by logic gates such as NOT, AND, OR..etc., in a sequential order.	Information / Data processing is carried out by quantum gates parallelly.
Calculations are deterministic: Repeated computation on the same input gives same output	Calculations are probabilistic: Multiple possible outputs for the same input. (Because of superposition and entanglement)
Low error rates when operated at room temperature.	High error rates at room temperature hence need to be kept ultra-cooled.
Operations are defined by Boolean algebra.	Operations are defined by linear algebra over Hilbert space and can be represented by unitary matrices with complex elements.
Circuit behavior is governed by classical physics.	Circuit behavior is governed by quantum mechanics.
Circuits are scalable and macroscopic technologies that are fast such as CMOS.	Circuits are not yet scalable and use microscopic technologies that are slow and fragile.

HILBERT SPACE

'An Hilbert space is an vector space which expands into an 'n' dimensional space which can be finite or infinite'

In quantum mechanics the state of a physical system is represented by a vector in a Hilbert space: a complex vector space with an inner product. The term "Hilbert space" is often reserved for an infinite-dimensional inner product space having the property that it is complete or closed.

(Note: Classical mechanics can describe physical properties of macroscopic objects, whereas quantum mechanics can describe physical properties at the microscopic scale. Many quantum mechanical phenomena are counter-intuitive and the Hilbert space model has been successfully in explaining and predicting the behaviors of quantum systems. Hilbert space model uses vectors and linear maps to give interpretation to explain the counter-intuitive phenomena.)

DIRAC'S BRA-KET NOTATIONS:

The Dirac's Bra-Ket notation is the concise and convenient way to describe the quantum states. It is often used to write linear algebra and its lengthy derivations in quantum mechanics.

Let a quantum state be represented by $|\alpha\rangle$ which is called a *Ket* or a *Ket vector* which is used to denote a *vector quantity in Hilbert space*. We say that a physical system is in quantum state $|\alpha\rangle$, where α is an Eigen value corresponding to an operator A that represents an physical measurement. If we have two distinct quantum states $|\alpha_1\rangle$ and $|\alpha_2\rangle$ then the following Ket is:

$$|\psi\rangle = c_1|\alpha_1\rangle + c_2|\alpha_2\rangle$$

Where, c_1 and c_2 are complex numbers and are also possible states for the systems.

Dirac also defined something called *Bra* or *Bra vector* represented by $\langle \beta |$ which does not belong to Ket space. For every Ket $|\alpha\rangle$ there exist a *dual of Ket called Bra* $\langle \alpha |$.

For example, if $c_1|\alpha_1\rangle + c_2|\alpha_2\rangle$ is a Ket vector, then the dual associated with its vector can be: $c_1^* \langle \alpha_1 | + c_2^* \langle \alpha_2 |$.

In general a ket vector is represented in matrix form with one column and 'n' number of rows

having 'n' number of elements as: $|\psi\rangle = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ \vdots \\ a_n \end{pmatrix}$.

Then, the Bra vector can be represented by row matrix as: $\langle \psi | = (a_1^* \ a_2^* \ a_3^* \ \dots \ a_n^*)$.

Here a_1^* is the complex conjugate of a_1 . It means, if $|\psi\rangle = \begin{bmatrix} 0 \\ i \end{bmatrix}$ then $\langle \psi | = [0 \ -i]$

Hence, a Bra vector is the Hermitian conjugate of Ket vector. i.e., $\langle \psi | = [|\psi\rangle]^\dagger$.

Also, If $|\psi\rangle = 2 + ib$ then $\langle \psi | = 2 - ib$

If $|\psi\rangle = 5 + 3i$ then $\langle \psi | = 5 - 3i$

If $|\psi\rangle = 2|\psi_1\rangle + 3|\psi_2\rangle$ then $\langle \psi | = 2\langle \psi_1 | + 3\langle \psi_2 |$

Hence, $\langle \psi | = [|\psi\rangle]^\dagger = [|\psi\rangle]^*{}^t$ i.e., a Bra vector is also a conjugate transpose of Ket vector.

Inner and Outer Product of Bra-Ket vectors:

Let ψ_i and ψ_j be two wave functions of the quantum states belonging to the same Hilbert space,

then the *inner product* can be written as: $\langle \psi_i | \psi_j \rangle = \delta_{ij} \begin{cases} 0 & \text{for } i \neq j \\ 1 & \text{for } i = j \end{cases}$.

Example: $\langle \psi_1 | \psi_1 \rangle = 1$ and $\langle \psi_1 | \psi_2 \rangle = 0$.

The inner product between the two states corresponds to the dot product between a Bra and a Ket vector as: $\langle \psi_i | \cdot | \psi_j \rangle \equiv \langle \psi_i | \psi_j \rangle$.

Similarly, the *Outer product* is written as: $|\psi_i\rangle \langle \psi_j|$

QUANTUM SUPERPOSITION

A quantum state is a superposition of classical states, written as a vector of amplitudes, to which we can apply either a measurement or a unitary operation.

Consider some physical system that can be in N different, mutually exclusive classical states. We call these states as $|0\rangle, |1\rangle, \dots, |N-1\rangle$. A pure quantum state (usually just called state) $|\psi\rangle$ is a superposition of classical states, written as:

$$|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle + \dots + \alpha_{N-1}|N-1\rangle$$

Her α_i is a complex number that is called amplitude of $|i\rangle$ in $|\psi\rangle$. A system in quantum state $|\psi\rangle$ is “in all classical states at the same time,” each state having a certain amplitude. It is in state $|0\rangle$ with amplitude α_0 , in state $|1\rangle$ with amplitude α_1 , and so on. Mathematically, the states $|0\rangle, |1\rangle, \dots, |N-1\rangle$ form an orthonormal basis of an N -dimensional Hilbert space. A quantum state $|\psi\rangle$ is a vector in this space, usually written as an N -dimensional column vector of its

amplitudes as : $|\psi\rangle = \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{N-1} \end{pmatrix}$

UNITARY EVOLUTION

Consider a quantum state, $|\psi\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle + \dots + \alpha_{N-1}|N-1\rangle$

Let the state changes to other form as: $|\phi\rangle = \beta_0|0\rangle + \beta_1|1\rangle + \dots + \beta_{N-1}|N-1\rangle$

Since, quantum mechanics only allows linear operations to be applied to quantum states. if we view a state like $|\psi\rangle$ as an N -dimensional vector, then applying an operation that changes $|\psi\rangle$ to $|\phi\rangle$ corresponds to multiplying $|\psi\rangle$ with an $N \times N$ complex-valued matrix U as:

$$U \begin{pmatrix} \alpha_0 \\ \alpha_1 \\ \vdots \\ \alpha_{N-1} \end{pmatrix} = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_{N-1} \end{pmatrix}$$

By linearity we have, $|\phi\rangle = U|\psi\rangle = U \sum_i \alpha_i |i\rangle$. Since measuring $|\phi\rangle$ should also give a probability distribution with one constraint that the sum of probability amplitudes must be 1.

$$\text{i.e., } \sum_{j=0}^{N-1} |\beta_j|^2 = 1$$

This implies that the operation U must preserve the norm of vectors, and hence must be a unitary transformation (often just called “a unitary”). A matrix U is unitary if its inverse U^{-1} equals its conjugate transpose U^* . This is equivalent to saying that U always maps a vector of norm 1 to a vector of norm 1. Because a unitary transformation always has an inverse, it follows that any (non-measuring) operation on quantum states must be reversible: by applying U^{-1} we can always “undo”

the action of U , and nothing is lost in the process. On the other hand, a measurement is clearly non-reversible, because we cannot reconstruct $|\psi\rangle$ from the observed classical state $|j\rangle$.

QUBIT

In classical computer the data is stored in a *bit*. At any instant of time the bit can take the value 0 or 1. The corresponding parameter in quantum computer is *qubit*. Qubit obeys the laws of quantum mechanics. The physical state of a qubit is denoted by wave function ψ .

A qubit can be in *superposition state* corresponding to logical states 0 and 1 just like a single photon. These states are represented as $|0\rangle$ and $|1\rangle$. The difference between bits and qubits is that a qubit can be in a state other than $|0\rangle$ or $|1\rangle$. It is also possible to form linear combinations of states, often called *superpositions*:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

The numbers α and β are complex numbers or real numbers. The state of a qubit is a vector in a two-dimensional complex vector space. The special states $|0\rangle$ and $|1\rangle$ are known as *computational basis states*, and form an *orthonormal basis for this vector space*.

According to quantum mechanics when we measure the information about the quantum state, the qubit will yield either the result 0, with probability $|\alpha|^2$, or the result 1, with probability $|\beta|^2$. Since the probabilities must sum to one: $|\alpha|^2 + |\beta|^2 = 1$ which means 'the qubit's state be normalized to length 1'. Thus, in general a qubit's state is a unit vector in a two-dimensional complex vector space.

Bell State: A qubit can exist in a continuum of states between $|0\rangle$ and $|1\rangle$ – until it is observed. When a qubit is measured, it only ever gives '0' or '1' as the measurement result – probabilistically.

Thus, before measurement, a qubit can be in the state: $\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$

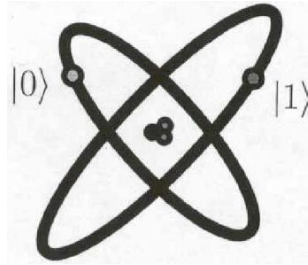
which, when measured, gives the result 0 fifty percent ($\left|\frac{1}{\sqrt{2}}\right|^2 = 0.5$ or 50%) of the time, and the result 1 fifty percent of the time. This state is called BELL STATE.

Physical Realization of Qubit:

Many different physical systems can be used to realize qubits, It could be:

1. Horizontal and vertical polarization of a single photon.
2. Right and left circular polarization of a single photon.
3. Up and down spin states of an electron.
4. Two energy states of an atom.
5. Two energy levels of a quantum dot.
6. Beam splitter modes of two particles.
7. Alignment of a nuclear spin in a uniform magnetic field

In the two spins states of an electron orbiting a single atom the electron can exist in either the so-called ‘ground’ or ‘excited’ states, which we’ll call $|0\rangle$ and $|1\rangle$, respectively as shown in the figure below. By shining light on the atom, with appropriate energy and for an appropriate length of time, it is possible to move the electron from the $|0\rangle$ state to the $|1\rangle$ state and vice versa. But more interestingly, by reducing the time we shine the light, an electron initially in the state $|0\rangle$ can be moved ‘halfway’ between $|0\rangle$ and $|1\rangle$.



GEOMETRICAL REPRESENTATION OF A QUBIT

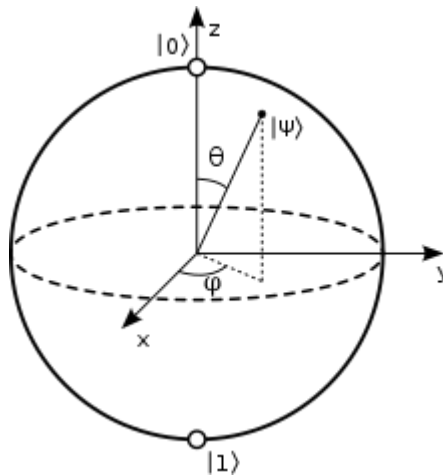
Consider the superposition state, $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, where $|\alpha|^2 + |\beta|^2 = 1$. Hence we may also write:

$$|\psi\rangle = e^{i\gamma} \left(\cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \right)$$

Where, θ, ϕ , and γ are real numbers. Ignoring $e^{i\gamma}$ since it has no observable effects, we can write:

$$|\psi\rangle = \cos \frac{\theta}{2} |0\rangle + e^{i\phi} \sin \frac{\theta}{2} |1\rangle \quad \rightarrow (1)$$

The numbers θ and ϕ define a point on the unit three-dimensional sphere as shown in the below figure. This sphere is often called the *Bloch sphere*; it provides a useful means of visualizing the state of a single qubit.



To *interpret* the superposition states considering the inherently probabilistic nature of observations on quantum systems, it is convenient to represent the state of a qubit in a ‘*Bloch sphere*’. Bloch sphere is a sphere of unit radius and the state of qubit is represented by a vector in this sphere as shown above figure:

Note, for $\phi = 0$ and $\theta = 0$, the state $|\psi\rangle$ corresponds to $|0\rangle$ and is along + z axis

For $\phi = 0$ and $\theta = 180^\circ$, the state $|\psi\rangle$ corresponds to $|1\rangle$ and is along - z axis

When $\theta = 90^\circ$, the vector is the x-y plane. For

$\phi = 90^\circ$, $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + i|1\rangle)$, is a superposition state along + y-axis

$\phi = -90^\circ$, $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$, is a superposition state along - y-axis

$\phi = 0^\circ$, $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$, is a superposition state along + x-axis

$\phi = 180^\circ$, $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$, is a superposition state along - x-axis

For classical computer, the two logical states 0 and 1 are represented by the poles of a sphere as show in the figure above. In contrast, the state of a qubit can be represented by any point on the sphere. Since there are infinite points on the sphere, a qubit in principle has more capacity to store information compared to a classical bit. *Bloch sphere represents the state of only one qubit, and there is no generalization of Bloch sphere for multiple qubits.*

QUANTUM ENTANGLEMENT

An important property quantum computing is *entanglement*, which refers to quantum correlations between different qubits. For instance, consider a 2-qubit register that is in the state as:

$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Such 2-qubit states are sometimes called *EPR-pairs* in honor of Einstein, Podolsky, and Rosen. Initially neither of the two qubits has a classical value $|0\rangle$ or $|1\rangle$. However, if we measure the first qubit and observe, say, a $|0\rangle$, then the whole state collapses to $|00\rangle$. Thus observing the first qubit immediately fixes the second. Since the two qubits that make up the register may be far apart, this example illustrates some of the *non-local effects* that quantum systems can exhibit. In general, a bipartite state $|\psi\rangle$ is called *entangled*.

Example: Consider a system of two electrons A and B with possible spins as up $|\uparrow\rangle$ and down $|\downarrow\rangle$ for both electrons. If these two electrons are not interacting with external agents and no measurement are being done on both electrons then they may have the following 4 possible combinations of spins:

$$|\uparrow_A\rangle|\uparrow_B\rangle \text{ or } |\uparrow_A\rangle|\downarrow_B\rangle \text{ or } |\downarrow_A\rangle|\uparrow_B\rangle \text{ or } |\downarrow_A\rangle|\downarrow_B\rangle$$

In superposition state, the system of two particles will be in the sum of all possible states as:

$$|\psi_{\text{initial}}\rangle = a.|\uparrow_A\rangle|\uparrow_B\rangle + b.|\uparrow_A\rangle|\downarrow_B\rangle + c.|\downarrow_A\rangle|\uparrow_B\rangle + d.|\downarrow_A\rangle|\downarrow_B\rangle$$

Where a, b, c, d are probability of finding the electrons with respective combinations of spins.

But when we start measuring the spin of any one of two electrons A and B then the system collapse to any one of the spin combination as: $|\psi_{\text{both}}\rangle = |\uparrow_A\rangle|\uparrow_B\rangle$ or $|\psi_{\text{both}}\rangle = |\downarrow_A\rangle|\downarrow_B\rangle$. Here the superposition states $|\uparrow_A\rangle|\downarrow_B\rangle$ and $|\downarrow_A\rangle|\uparrow_B\rangle$ cannot be measured.

If such system exists for infinite times i.e., if 'n' such identical system exists, then:

a^2n times the system will be in state $|\uparrow_A\rangle|\uparrow_B\rangle$, where a^2 is the *probability of finding the system in state* $|\uparrow_A\rangle|\uparrow_B\rangle$

d^2n times the system will be in state $|\downarrow_A\rangle|\downarrow_B\rangle$, where d^2 is the *probability of finding the system in state* $|\downarrow_A\rangle|\downarrow_B\rangle$.

Since, a and d can be complex then $|a|^2, |d|^2$.. are *probabilities or probability amplitudes*.

But if the system of two electrons are in the entangled state, then the probability distribution of spin i.e., likelihood of finding the electron B in a particular spin would change every time when we interact with electron A. i.e., the whole system is disturbed.

Hence we make use of **Bell states** as: $\frac{1}{\sqrt{2}}|\uparrow_A\rangle|\downarrow_B\rangle + \frac{1}{\sqrt{2}}|\downarrow_A\rangle|\uparrow_B\rangle$

This means, the probability of $|\uparrow_A\rangle|\uparrow_B\rangle$ and $|\downarrow_A\rangle|\downarrow_B\rangle$ is zero.

Then, before measuring spin of A, with Bell state, the spin of B is 50% $|\uparrow_B\rangle$ and 50% $|\downarrow_B\rangle$.

But, after measuring spin of A as $|\uparrow_A\rangle$ then B must have 100% $|\downarrow_B\rangle$. OR if A is $|\downarrow_A\rangle$ then B must have 100% $|\uparrow_B\rangle$.

Thus, the system of two electrons collapsed from 4 possible combinations to any of 2.

Note: In the above example, in the ENTANGLED state, the probability of electron B with specific spin say $|\uparrow_B\rangle$ is different before measuring A spin, after measuring A as $|\uparrow_A\rangle$ and after measuring A as $|\downarrow_A\rangle$.

QUANTUM GATE (HADAMARD GATE)

A unitary that acts on a small number of qubits (say, at most 3) is often called a gate, in analogy to classical logic gates like AND, OR, and NOT. The Pauli matrices are examples of 1-qubit gates.

For example, the bitflip gate X (a.k.a. NOT-gate) negates the bit in the computational basis, i.e., it swaps $|0\rangle$ and $|1\rangle$. The phase flip gate Z puts a $-$ in front of $|1\rangle$. Represented as 2×2 unitary

matrices, these are: $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

The most important 1-qubit gate is the **Hadamard transform**, specified by: $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

If we apply H to initial state $|0\rangle$ and then measure, we have equal probability of observing $|0\rangle$ or $|1\rangle$. Similarly, applying H to $|1\rangle$ and observing gives equal probability of $|0\rangle$ or $|1\rangle$.

However, if we apply H to the superposition $\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$ then we obtain:

$$H\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) = \frac{1}{\sqrt{2}}H|0\rangle + \frac{1}{\sqrt{2}}H|1\rangle = \frac{1}{2}(|0\rangle + |1\rangle) + \frac{1}{2}(|0\rangle - |1\rangle) = |0\rangle$$

The positive and negative amplitudes for $|1\rangle$ have canceled each other out. This effect is called ***interference***.
